Hybrid Bengali Character Recognition Using a Fast 14M Residual LeNet-Style Deep Neural Network

Author: Sayantan Roy
Department of Computer Science and Engineering
Government College of Engineering and Ceramic Technology (GCECT), Kolkata, India

Repository: https://github.com/sayantanr/hybrid-bengali-character-recognition

---

Abstract

Optical Character Recognition (OCR) for Bengali script remains a challenging problem due to the large number of visually similar characters, variations in handwriting styles, and complex structural patterns inherent in Indic scripts. This paper presents a novel Hybrid Bengali Character Recognition framework based on a Fast 14 Million Parameter Residual LeNet-Style Convolutional Neural Network integrated with a lightweight feature-mixing module.

The proposed architecture combines classical LeNet-inspired feature extraction, deep residual learning, and modern multilayer feature mixing to achieve high recognition accuracy while maintaining computational efficiency. Experiments were conducted on a custom Bengali handwritten character dataset consisting of 15,000 images distributed across 50 classes.

The model achieved a peak test accuracy of 98.57%, macro F1-score of 0.99, and weighted F1-score of 0.99, demonstrating excellent generalization capability across all character classes. Results indicate that hybridization of traditional CNN principles with residual learning and feature mixing offers an effective solution for Bengali handwritten character recognition.

Keywords: Bengali OCR, Character Recognition, Deep Learning, Residual CNN, LeNet, Computer Vision, Handwritten Character Recognition, Indic Language Processing

---

1. Introduction

Handwritten character recognition has been a major research area within computer vision and pattern recognition for decades. While considerable progress has been achieved for English and Latin scripts, Indic languages continue to pose unique challenges because of:

- Large character sets
- Complex stroke structures
- High intra-class variability
- Similar-looking characters
- Writing style diversity

Bengali is one of the most widely spoken languages in the world, yet comparatively fewer high-performance OCR systems have been developed specifically for Bengali handwritten characters.

Recent advances in deep learning have shown that Convolutional Neural Networks (CNNs) can automatically learn hierarchical representations directly from image data. However, many existing architectures either suffer from excessive computational complexity or are not optimized for Bengali character morphology.

This work proposes a Hybrid Fast14M Architecture that integrates:

1. LeNet-inspired convolutional feature extraction
2. Residual learning mechanisms
3. Lightweight feature mixing
4. Efficient classification layers

The objective is to maximize recognition performance while maintaining practical training speed and deployment feasibility.

---

2. Related Work

Early Bengali OCR systems relied heavily on:

- Template matching
- Histogram features
- Zoning methods
- Statistical classifiers

Subsequently, machine learning approaches such as:

- Support Vector Machines
- Random Forests
- k-Nearest Neighbors

were introduced.

The deep learning era brought:

- LeNet
- AlexNet
- VGG
- ResNet
- EfficientNet
- Vision Transformers

Although transformer-based architectures have achieved state-of-the-art performance in several vision tasks, CNN-based methods remain attractive for medium-scale OCR applications due to lower computational requirements and faster inference.

This research explores a hybrid CNN paradigm that leverages residual learning while preserving the efficiency of classical convolutional architectures.

---

3. Dataset Description

The experimental dataset contains:

Property	Value
Total Classes	50
Training Images	12,000
Testing Images	3,000
Total Images	15,000
Image Resolution	64 × 64
Type	Handwritten Bengali Characters

Each class contains approximately:

- 240 training samples
- 60 testing samples

ensuring balanced class representation.

---

4. Proposed Hybrid Architecture

4.1 Overview

The proposed network follows:

Input Image (64×64) ↓ LeNet-Style Stem ↓ Residual CNN Stages ↓ Adaptive Global Pooling ↓ MLP Mixer ↓ Classifier

---

4.2 LeNet-Inspired Feature Stem

The initial stage employs:

- Convolution (3→32)
- Batch Normalization
- ReLU
- Max Pooling

followed by:

- Convolution (32→64)
- Batch Normalization
- ReLU
- Max Pooling

This stage captures low-level structural features such as:

- edges
- curves
- stroke intersections

which are particularly important in Bengali script.

4.3 Residual Learning Module

The architecture utilizes multiple residual blocks consisting of:

- 3×3 convolutions
- Batch normalization
- SiLU activation
- Identity skip connections

Residual learning alleviates:

- vanishing gradients
- optimization difficulties
- degradation in deeper networks

allowing efficient learning of character-specific patterns.

4.4 Feature Mixer

After convolutional feature extraction, a lightweight feature mixing module performs:

512 → 2048 → 512

transformations using:

- Fully connected layers
- SiLU activations
- Dropout regularization

This stage improves global feature interactions and class separability.

4.5 Classification Layer

The final classifier maps learned representations into:

50 Bengali character classes

using a fully connected output layer with softmax-based classification.

---

5. Experimental Setup

Hardware

Training was performed using:

- NVIDIA CUDA GPU
- PyTorch Framework

Hyperparameters

Parameter	Value
Epochs	20
Batch Size	256
Learning Rate	0.002
Optimizer	AdamW
Weight Decay	1e-4
Label Smoothing	0.05
Scheduler	OneCycleLR
Input Size	64×64

Total Parameters

14,474,866 trainable parameters

(~1.45 Crore Parameters)

---

6. Training Performance

The learning curve demonstrated rapid convergence.

Epoch	Test Accuracy
1	3.33%
5	91.80%
10	97.70%
15	98.20%
20	98.57%

The network achieved over 90% accuracy within only five epochs, indicating highly efficient feature learning.

---

7. Results

Best Model

Fast14M_Residual_LeNetStyle_CNN

Performance Metrics

Metric	Value
Accuracy	98.57%
Macro Precision	0.99
Macro Recall	0.99
Macro F1-Score	0.99
Weighted F1-Score	0.99
Training Time	505.56 seconds

Class-Level Performance

The majority of classes achieved:

- Precision $\geq$ 0.98
- Recall $\geq$ 0.98
- F1-score $\geq$ 0.98

Several classes achieved perfect classification:

- Class 181
- Class 182
- Class 200

- Class 206
- Class 210
- Class 215
- Class 217
- Class 218
- Class 220

with Precision = Recall = F1 = 1.00.

---

8. Confusion Matrix Analysis

The confusion matrix reveals that:

- Most classes exhibit near-perfect diagonal dominance.
- Misclassifications are sparse.
- Errors occur primarily between visually similar Bengali characters.

Examples include:

- 172 ↔ 173
- 174 ↔ 175
- 176 ↔ 177
- 202 and neighboring classes

These confusions are expected because many Bengali handwritten characters share structural similarities.

Nevertheless, the error rate remains extremely low.

---

9. Discussion

The obtained accuracy of 98.57% demonstrates the effectiveness of combining:

- Classical LeNet concepts
- Residual learning
- Modern optimization techniques
- Feature mixing mechanisms

The architecture provides several advantages:

High Accuracy

Achieves near state-of-the-art performance on the experimental dataset.

Fast Convergence

Exceeds 90% accuracy within five epochs.

Computational Efficiency

Requires significantly fewer resources than large transformer-based models.

Scalability

Can be extended to:

- Bengali numerals
- Bengali compound characters
- Bengali word recognition
- Full OCR pipelines

---

10. Limitations

Current limitations include:

1. Evaluation limited to isolated character recognition.
2. Compound characters were not included.
3. Dataset size remains moderate compared to industrial OCR corpora.
4. Transformer-based comparisons were not performed.

Future work should address these limitations.

---

11. Future Research Directions

Potential extensions include:

- Vision Transformer integration
- CNN–Transformer hybrids
- Self-supervised Bengali representation learning
- Few-shot Bengali OCR
- Bengali compound character recognition
- Bengali word-level OCR
- Document-level OCR systems
- Multilingual Indic OCR frameworks

---

12. Conclusion

This paper introduced a Hybrid Bengali Character Recognition framework based on a Fast 14M Residual LeNet-Style Deep Neural Network. The proposed model successfully combines traditional convolutional feature extraction with residual learning and feature mixing to achieve highly accurate handwritten Bengali character recognition.

Experimental evaluation on a 50-class Bengali handwritten dataset containing 15,000 images demonstrated a peak classification accuracy of 98.57%, macro F1-score of 0.99, and excellent class-wise performance. The results indicate that the proposed architecture provides an effective balance between computational efficiency and recognition accuracy.

The presented framework offers a promising foundation for future Bengali OCR systems and highlights the continued relevance of optimized CNN architectures for Indic language processing tasks.

References

[1] LeCun, Y., Bottou, L., Bengio, Y., Haffner, P. "Gradient-Based Learning Applied to Document Recognition."

[2] He, K., Zhang, X., Ren, S., Sun, J. "Deep Residual Learning for Image Recognition."

[3] Dosovitskiy, A., et al. "An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale."

[4] Krizhevsky, A., Sutskever, I., Hinton, G. "ImageNet Classification with Deep Convolutional Neural Networks."

[5] Roy, S. Hybrid Bengali Character Recognition Repository. GitHub.